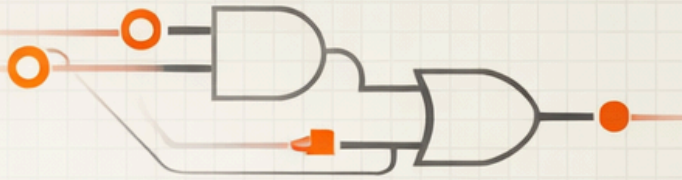

Lógica para Ciência da Computação

Uma abordagem ativa utilizando a Peer Instruction

Esdras L. Bispo Jr.

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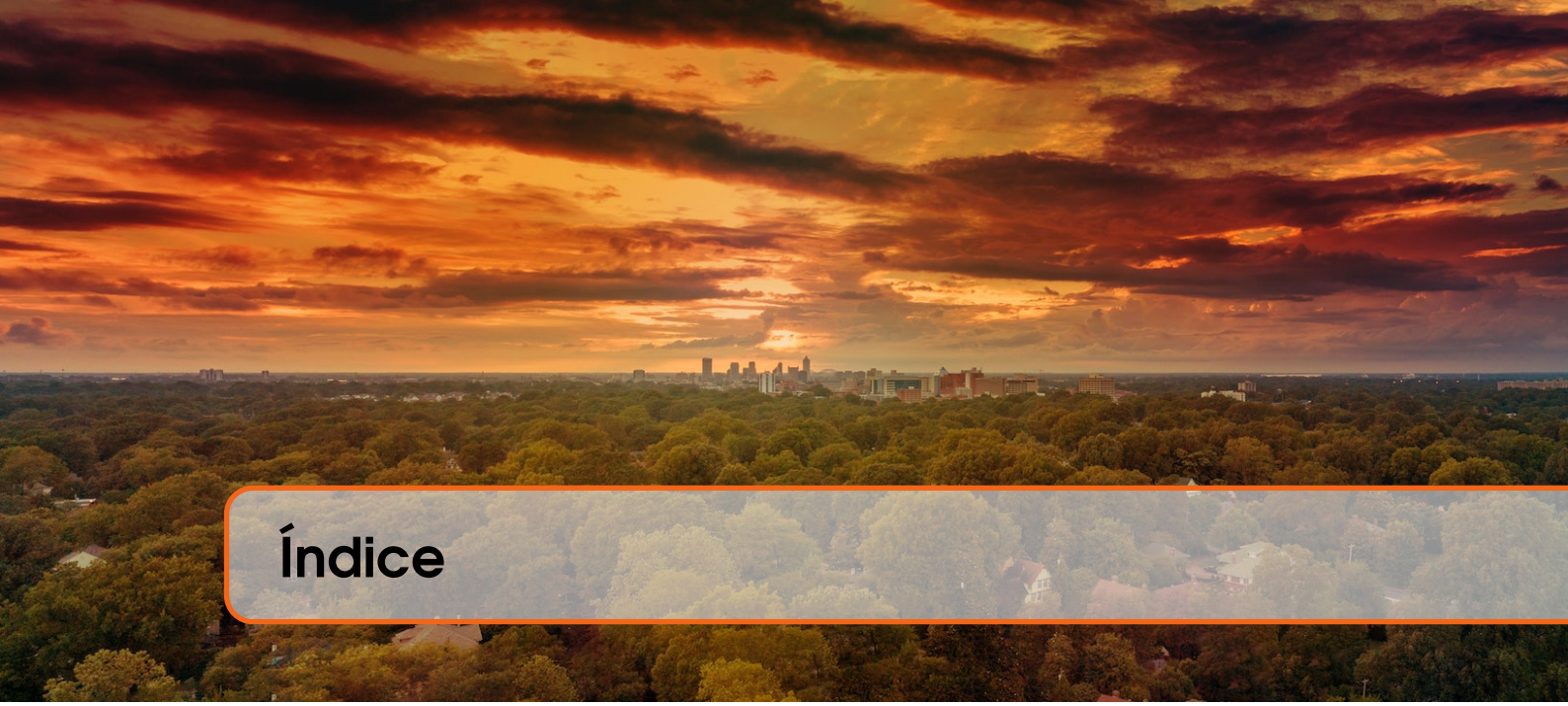
First printing, March 2022



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1. Sectioning Examples

1.1 Section Title

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1.1.1 Subsection Title

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Unnumbered Section

Unnumbered Subsection

Unnumbered Subsubsection

2. In-text Element Examples

2.1 Referencing Publications

This statement requires citation [1]; this one is more specific [2, página 162].

2.2 Link Examples

This is a URL link: LaTeX Templates. This is an email link: example@example.com. This is a monospaced URL link: `https://www.LaTeXTemplates.com`.

2.3 Lists

Lists are useful to present information in a concise and/or ordered way.

2.3.1 Numbered List

1. First numbered item
    a. First indented numbered item
    b. Second indented numbered item
        i. First second-level indented numbered item
2. Second numbered item
3. Third numbered item

2.3.2 Bullet Point List

- First bullet point item
    - First indented bullet point item
    - Second indented bullet point item
        - First second-level indented bullet point item
- Second bullet point item
- Third bullet point item

2.3.3 Descriptions and Definitions

Name Description
Word Definition
Comment Elaboration

2.4 International Support

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ÀÁÂÄÃÅÈÉÊËÌÍÎÏÒÓÔÖÕØÙÚÛÜŸÝÑ
ßÇŒÆČŠŽ

2.5 Ligatures

fi fj fl ffl ffi Ty

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3. Mathematics

3.1 Theorems

3.1.1 Several equations

This is a theorem consisting of several equations.

Theorem 3.1 — Name of the theorem. In $E = \mathbb{R}^n$ all norms are equivalent. It has the properties:

$$\big|\,||\mathbf{x}|| - ||\mathbf{y}||\,\big| \leq ||\mathbf{x} - \mathbf{y}|| \tag{3.1}$$

$$||\sum_{i=1}^{n} \mathbf{x}_i|| \leq \sum_{i=1}^{n} ||\mathbf{x}_i|| \quad \text{where } n \text{ is a finite integer} \tag{3.2}$$

3.1.2 Single Line

This is a theorem consisting of just one line.

Theorem 3.2 A set $\mathscr{D}(G)$ in dense in $L^2(G)$, $|\cdot|_0$.

3.2 Definitions

A definition can be mathematical or it could define a concept.

Definition 3.1 — Definition name. Given a vector space E, a norm on E is an application, denoted $||\cdot||$, E in $\mathbb{R}^+ = [0, +\infty[$ such that:

$$||\mathbf{x}|| = 0 \;\Rightarrow\; \mathbf{x} = \mathbf{0} \tag{3.3}$$

$$||\lambda \mathbf{x}|| = |\lambda| \cdot ||\mathbf{x}|| \tag{3.4}$$

$$||\mathbf{x} + \mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}|| \tag{3.5}$$

3.3 Notations

- **Notation 3.1** Given an open subset G of $\mathbb{R}^n$, the set of functions φ are:
  1. Bounded support G;
  2. Infinitely differentiable;

a vector space is denoted by $\mathscr{D}(G)$.

3.4 Remarks

This is an example of a remark.

R The concepts presented here are now in conventional employment in mathematics. Vector spaces are taken over the field $\mathbb{K} = \mathbb{R}$, however, established properties are easily extended to $\mathbb{K} = \mathbb{C}$.

3.5 Corollaries

Corollary 3.1 — Corollary name. The concepts presented here are now in conventional employment in mathematics. Vector spaces are taken over the field $\mathbb{K} = \mathbb{R}$, however, established properties are easily extended to $\mathbb{K} = \mathbb{C}$.

3.6 Propositions

3.6.1 Several equations

Proposition 3.1 — Proposition name. It has the properties:

$$\big|||\mathbf{x}|| - ||\mathbf{y}||\big| \leq ||\mathbf{x} - \mathbf{y}|| \tag{3.6}$$

$$||\sum_{i=1}^{n} \mathbf{x}_i|| \leq \sum_{i=1}^{n} ||\mathbf{x}_i|| \quad \text{where } n \text{ is a finite integer} \tag{3.7}$$

3.6.2 Single Line

Proposition 3.2 Let $f, g \in L^2(G)$; if $\forall \varphi \in \mathscr{D}(G)$, $(f, \varphi)_0 = (g, \varphi)_0$ then $f = g$.

3.7 Examples

3.7.1 Equation Example

■ **Example 3.1** Let $G = \{x \in \mathbb{R}^2 : |x| < 3\}$ and denoted by: $x^0 = (1, 1)$; consider the function:

$$f(x) = \begin{cases} \mathrm{e}^{|x|} & \text{si } |x - x^0| \leq 1/2 \\ 0 & \text{si } |x - x^0| > 1/2 \end{cases} \tag{3.8}$$

The function f has bounded support, we can take $A = \{x \in \mathbb{R}^2 : |x - x^0| \leq 1/2 + \varepsilon\}$ for all $\varepsilon \in]0; 5/2 - \sqrt{2}[$. ■

3.7.2 Text Example

■ **Example 3.2 — Example name.** Aliquam arcu turpis, ultrices sed luctus ac, vehicula id metus. Morbi eu feugiat velit, et tempus augue. Proin ac mattis tortor. Donec tincidunt, ante rhoncus luctus semper, arcu lorem lobortis justo, nec convallis ante quam quis lectus. Aenean tincidunt sodales massa, et hendrerit tellus mattis ac. Sed non pretium nibh. Donec cursus maximus luctus. Vivamus lobortis eros et massa porta porttitor. ■

3.8 Exercises

Exercise 3.1 This is a good place to ask a question to test learning progress or further cement ideas into students' minds. ■

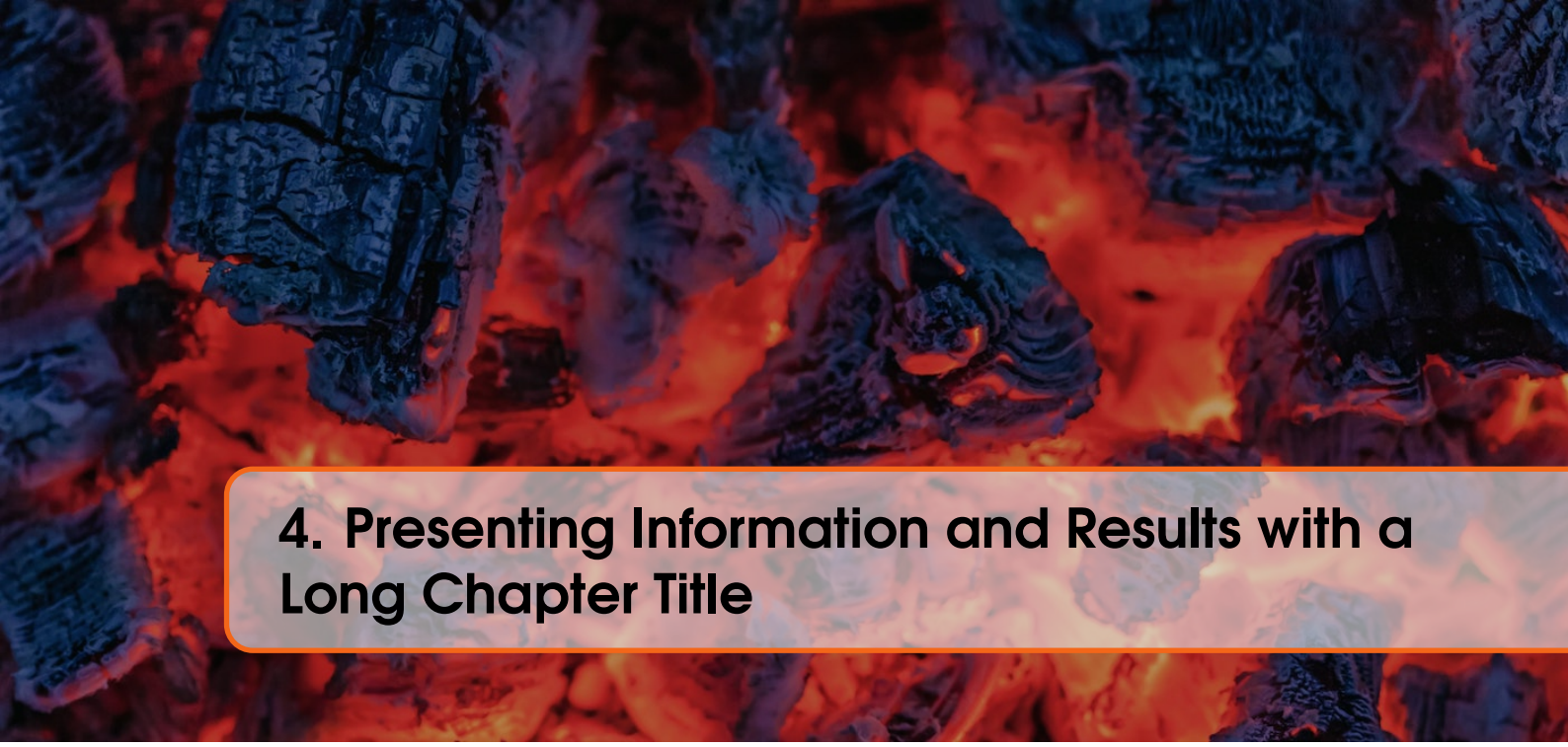
3.9 Problems

Problem 3.1 What is the average airspeed velocity of an unladen swallow?

3.10 Vocabulary

Define a word to improve a students' vocabulary.

- **Vocabulary 3.1 — Word.** Definition of word.



4. Presenting Information and Results with a Long Chapter Title

4.1 Table

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Treatments	**Response 1**	**Response 2**
Treatment 1	0.0003262	0.562
Treatment 2	0.0015681	0.910
Treatment 3	0.0009271	0.296

Tabela 4.1: Table caption.

Referencing Table 4.1 in-text using its label.

4.2 Figure

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Figura 4.1: Figure caption.

Referencing Figure 4.1 in-text using its label.

Treatments	**Response 1**	**Response 2**
Treatment 1	0.0003262	0.562
Treatment 2	0.0015681	0.910
Treatment 3	0.0009271	0.296

Tabela 4.2: Floating table.

Figura 4.2: Floating figure.

Bibliography

Articles

[1] A. B. Jones e J. M. Smith. “Article Title”. Em: *Journal title* 13.52 (mar. de 2022), páginas 123–456. DOI: 10.1038/s41586-021-03616-x (ver página 13).

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[2] J. M. Smith e A. B. Jones. *Book Title*. 7th. Publisher, 2021 (ver página 13).

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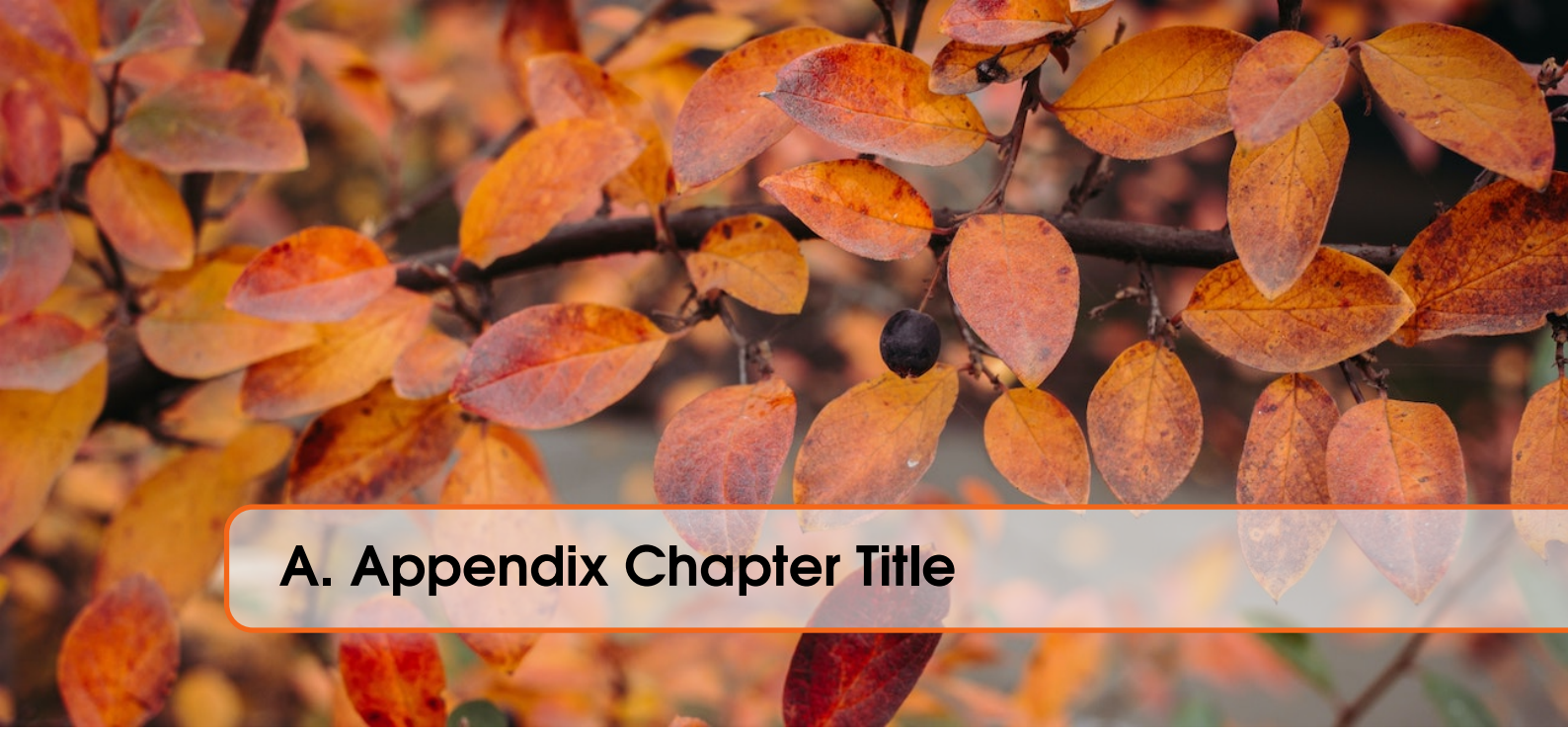

A. Appendix Chapter Title

A.1 Appendix Section Title

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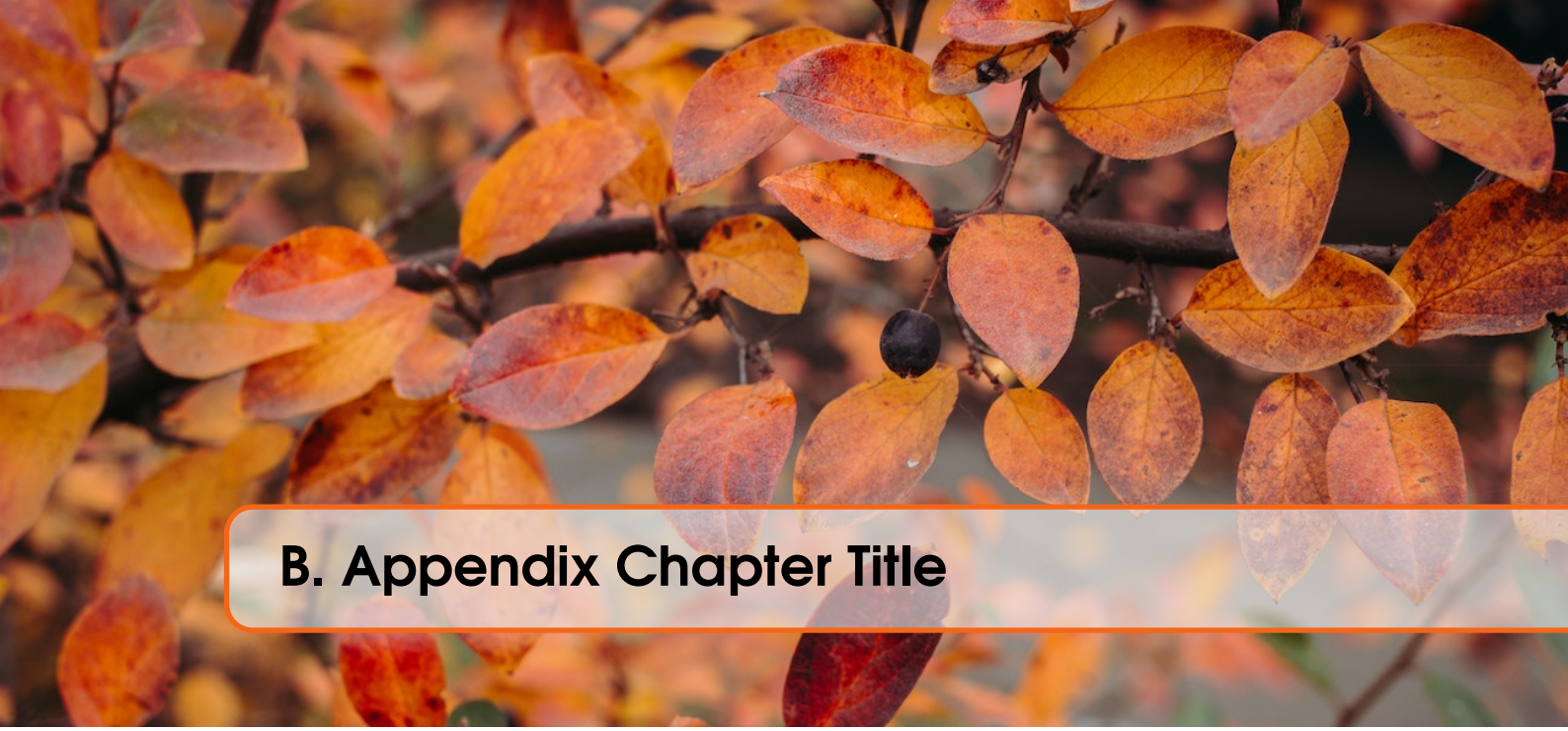

B. Appendix Chapter Title

B.1 Appendix Section Title

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